

ColorIt

Local full-movie AI colorization - with one command



Beautiful restoration **exists**

Most people cannot **afford** it

Traditional

- Professional team
- Manual coloring
- Weeks of effort

ColorIt

- One person
- One command
- Few hours

Restoration Access at Three Scales



Individuals coloring
favorite films and
personal clip



Restoration experts
generating an AI
first pass



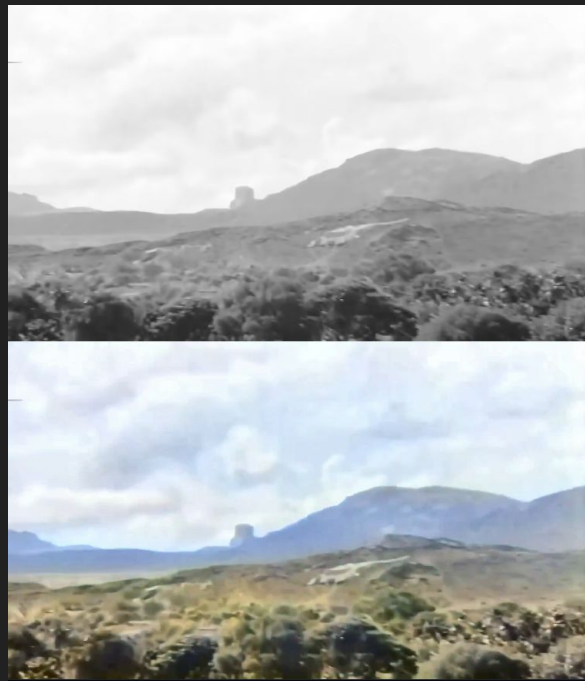
Studios and rights
holders exploring
back catalogs

A Practical Product

Usage

```
darksca@Darks-Mac-Studio:~/ColorIt$ uv run colorit colorize-movie
--input demo.mp4 --overwrite
Movie: /Users/darksca/ColorIt/demo.mp4
Output: /Users/darksca/ColorIt/demo_color.mp4
Config: /Users/darksca/ColorIt/configs/full_movie.yaml
Scene threshold: 0.60
Config: /Users/darksca/ColorIt/configs/full_movie.yaml
Movie: /Users/darksca/ColorIt/demo.mp4
Scene threshold: 0.6
Movie duration seconds: 60.001
Scene detection: 100%|█| 1500/1500 [00:01<00:00, 1252.55frame/s]
Scene manifest written to /Users/darksca/ColorIt/data/manifests/demo_ec388e7b_t060.json
Scene units: 1
Movie: /Users/darksca/ColorIt/demo.mp4
Scene manifest: /Users/darksca/ColorIt/data/manifests/demo_ec388e7b_t060.json
Batch scene count: 1
Resume mode: False
Scene extraction: mezzanine: 100%|█| 1500/1500 [00:05<00:00, 265
Encoded scene mezzanine: source_mezzanine.mp4
Scene extraction: frame split: 100%|█| 1500/1500 [00:07<00:00, 2
Frame-exact scene clips written from mezzanine: 1
Extracted scene clip: scene_0001.mp4
CLAHE scene_0001: 79%|█| 1178/1500 [00:08<00:02, 143.95frame/s]
```

Results

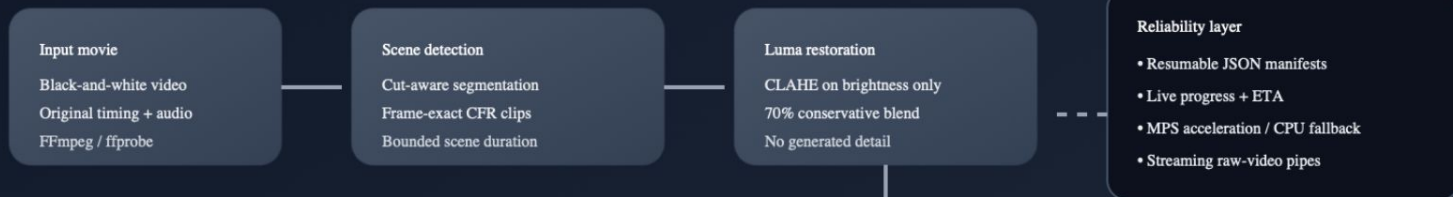


Two AI Perspectives -> One Video

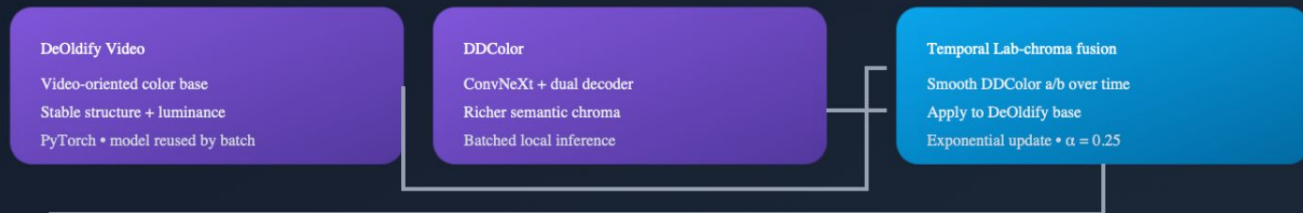
ColorIt: an automatic movie colorization pipeline

Scene-aware dual-model inference • temporal chroma fusion • timing-safe delivery

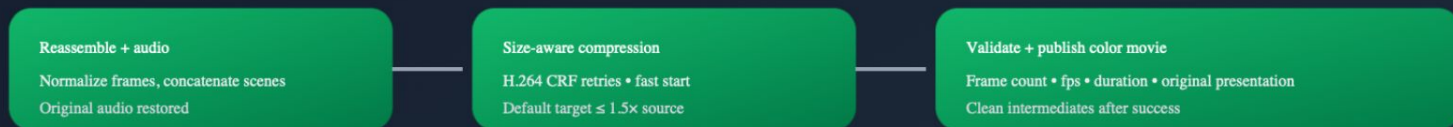
1 · PREPARE



2 · INFER + FUSE EACH SCENE



3 · DELIVER



Color is Inferred; Continuity is Engineered

1. Grayscale input does not contain chroma: learned visual semantics are essential
2. DeOldify and DDColor have complementary strengths
3. ColorIt isolates luminance and temporally smooths DDColor's Lab a/b prediction before applying it to the DeOldify base
4. Scene boundaries, frame-exact assembly, and validation operationalize the AI at movie scale

DeOldify
base



DDColor
chroma



Temporally
fused
result

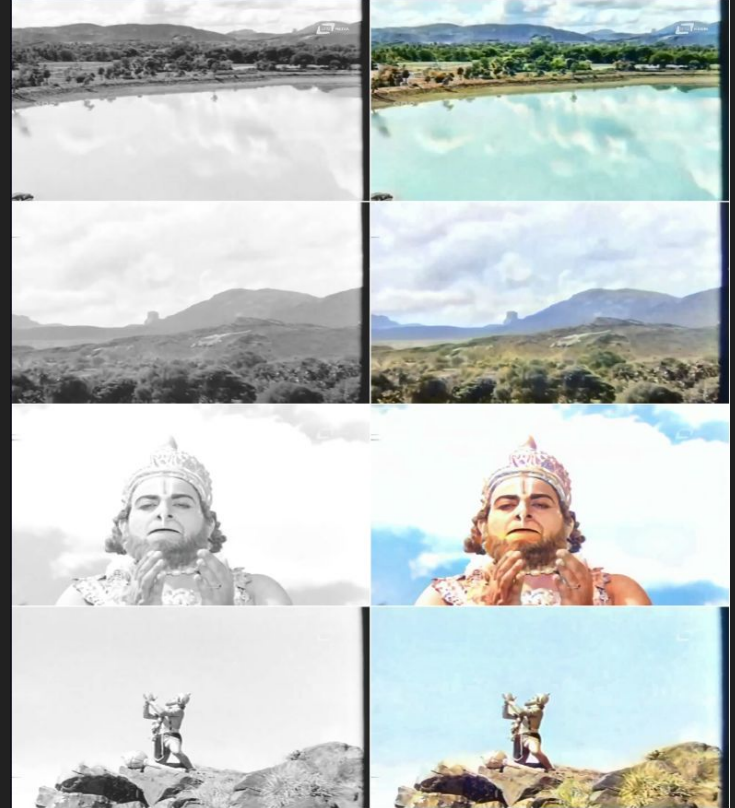
Working Result and Validation

Colorlt Quality

Strong: sky, water, forest,
mountain

Reasonable: actor skin tone

Needs work: vivid costume
palette and cross-cut identity



Future Work



Train

temporally conditioned colorization with actor, skin-tone, costume, and palette context

Upscale

to 1080p/4K detail using generative reconstruction

Translate

into other languages and integrate ASR, subtitles, and speech-aware lip synchronization

Controls

for experts who want historical accuracy

Setup

Follow README instructions in <https://github.com/Vineet-Vinod/ColorIt>

Colored Clip Demo

Video at https://youtu.be/_HlflRmK9qI

Demo clip taken from Ramanjaneya Yuddha movie uploaded by SRS media (00:58:13 to 00:59:13) - <https://youtu.be/qIv9D334AVg>